ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Coordination Chemistry

English Medium

EXERCISE # O-1

SINGLE OPTION CORRECT :

Double salt and complex compound

1. Some salts although containing two different metallic elements give test for one of them in solution. Such salts are

(A) complex salt (B) double salt (C) normal salt (D) none of these **CC0001**

2. Aqueous solution of $FeSO_4$ gives tests for both Fe^{2+} and SO_4^{2-} but after addition of excess of KCN, solution ceases to give test for Fe^{2+}. This is due to the formation of

(A) the double salt $FeSO_4.2KCN.6H_2O$ (B) $Fe(CN)_3$

(C) the complex ion $[Fe(CN)_6]^{4-}$ (D) the complex ion $[Fe(CN)_6]^{3-}$ **CC0002**

Werner theory

3. Consider the following statements:

According the Werner's theory.

(a) Ligands are connected to the metal ions by covalent bonds.

(b) Secondary valencies have directional properties

(c) Secondary valencies are non-ionisable

Of these statements:

(A) a, b and c are correct (B) b and c are correct

(C) a and c are correct (D) a and b are correct **CC0003**

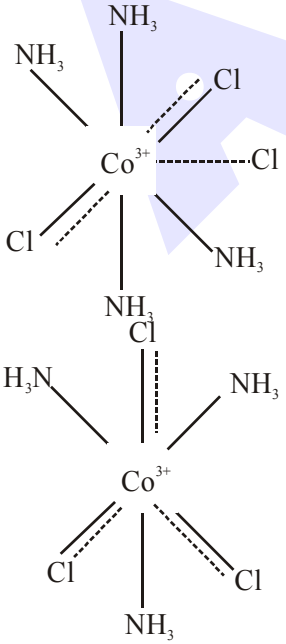
4. A complex of platinum, ammonia and chloride produces four ions per molecule in the solution. The structure consistent with the observation is:

(A) $[Pt(NH_3)_4]Cl_4$ (B) $[Pt(NH_3)_2Cl_4]$ (C) $[Pt(NH_3)_5Cl]Cl_3$ (D) $[Pt(NH_3)_4Cl_2]Cl_2$

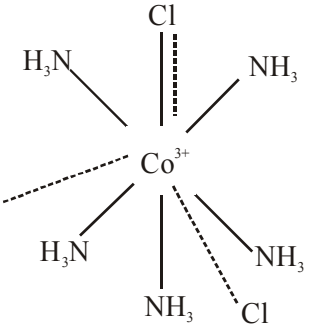
CC0004

5. Which of the following Werner's complex has least electrical conductivity ?

(A)

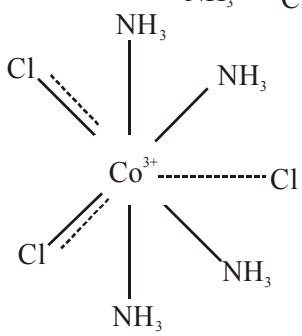


(B)



(C)

(D)



CC0005

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Classification of ligand

6. How many $EDTA^{-4}$ molecules are required to make an octahedral complex with a Ca^{2+} ion ?
(A) Six (B) Three (C) One (D) Two **CC0006**

7. π-bonding is not involved in :
(A) Ferrocene (B) Dibenzene chromium (C) Zeise's salt (D) Grignard reagent
CC0007

8. Which of the following is not considered as an organometallic compound ?
(A) Ferrocene (B) Cis-platin (C) Ziese's salt (D) Grignard reagent
CC0008

9. Diethylene triamine is:
(A) Chelating agent (B) Polydentate ligand
(C) Tridentate ligand (D) All of these **CC0009**

10. Which of the following species is not expected to be a ligand
(A) NO^+ (B) NH_4^+ (C) $NH_2-NH_3^+$ (D) CO **CC0010**

11. The disodium salt of ethylene diamine tetraacetic acid can be used to estimate the following ion(s) in the aqueous solution
(A) Mg^{2+} ion (B) Ca^{2+} ion (C) Na^+ ion (D) both Mg^{2+} and Ca^{2+}
CC0011

12. Which of the following ligand does not act as bidentate ligand
(A) dipy (B) dien (C) gly^- (D) dmg^- **CC0012**

Synergic bonding

13. Which of the following order is correct for the IR vibrational frequency of CO.
(A) $[Fe(CO)_4]^{2-} < [Co(CO)_4]^- < [Ni(CO)_4]$ (B) $[Fe(CO)_4]^{2-} > [Co(CO)_4]^- > [Ni(CO)_4]$
(C) $[Fe(CO)_4]^{2-} > [Co(CO)_4]^- < [Ni(CO)_4]$ (D) $[Fe(CO)_4]^{2-} < [Co(CO)_4]^- > [Ni(CO)_4]$
CC0013

14. In the isoelectronic series of metal carbonyl, the C–O bond strength is expected to increase in the order.
(A) $[Mn(CO)_6]^+ < [Cr(CO)_6] < [V(CO)_6]^-$ (B) $[V(CO)_6]^- < [Cr(CO)_6] < [Mn(CO)_6]^+$
(C) $[V(CO)_6]^- < [Mn(CO)_6]^+ < [Cr(CO)_6]$ (D) $[Cr(CO)_6] < [Mn(CO)_6]^+ < [V(CO)_6]^-$
CC0014

15. Which of the following has higher stretching frequency for C–O bond -
(A) $[Ni(CO)_3PF_3]$ (B) $[Ni(CO)_3(PMe_3)]$
(C) Both have equal stretching frequency (D) None of these **CC0015**

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16. Which of the following has higher multiple bond character in M–C bond -

(A) $[Ni(CO)_4]$

(B) $[Co(CO)_4]^-$

(C) $[Fe(CO)_4]^{2-}$

(D) (B) and (C) both have equal multiple bond character in M–C bond **CC0016**

17. The V–C distance in $V(CO)_6$ and $[V(CO)_6]^-$ are respectively (in pm)-

(A) 200, 200 (B) 193, 200 (C) 200, 193 (D) 193, 193 **CC0017**

Co-ordination number and E.A.N.

18. Among the following complexes which can act as oxidising agent.

(A) $[Mn(CO)_6]$ (B) $[Mn(CO)_6]^+$ (C) $[Mn(CO)_5]^-$ (D) $[V(CO)_6]$ **CC0018**

19. Which of the following statement is correct regarding the compound " $[(CO)_3Fe(CO)_3Fe(CO)_3]$".

(A) The d_{C-O} (bridging) is greater than d_{C-O} (terminal)

(B) The bond order of bridging C – O bond is greater than that of terminal C – O bond

(C) The E.A.N. value of each Fe-atom is 35

(D) The oxidation state of Fe in this complex is (–I)

CC0019

20. How many π electrons are donated by $C_5H_5^-$ ligand -

(A) 2 (B) 4 (C) 5 (D) 6 **CC0020**

21. Effective atomic number of $Co(CO)_4$ is 35, hence it is less stable. It attains stability by

(A) Oxidation of Co (B) Reduction of Co

(C) Dimerization (D) Both (B) & (C) **CC0021**

22. In the complex $Fe(CO)_x$, the value of x is:

(A) 3 (B) 4 (C) 5 (D) 6 **CC0022**

23. The EAN of platinum in potassium hexachloroplatinate (IV) is:

(A) 46 (B) 86 (C) 36 (D) 84 **CC0023**

24. The EAN of metal atoms in $Fe(CO)_2(NO)_2$ and $Co_2(CO)_8$ respectively are

(A) 34, 35 (B) 34, 36 (C) 36, 36 (D) 36, 35 **CC0024**

Naming of complex compound

25. The IUPAC name for the coordination compound $Ba[BrF_4]_2$ is

(A) Barium tetrafluorobromate (V) (B) Barium tetrafluorobromate (III)

(C) Barium bis (tetrafluorobromate) (III) (D) none of these **CC0025**

26. The number of ions formed, when bis (ethane-1,2-diamine) copper (II) sulphate is dissolved in water will be

(A) 1 (B) 2 (C) 3 (D) 4 **CC0026**

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27. The IUPAC name of the Wilkinson's catalyst [Rh Cl (P Ph_3)$_3$] is

(A) Chloridotris(triphenylphosphine)rhodium(I)

(B) Chloridotris(triphenylphosphine)rhodium(IV)

(C) Chloridotris(triphenylphosphine)rhodium(0)

(D) Chloridotris(triphenylphosphine)rhodium(VI) **CC0027**

28. The formula for the compound tris (ethane-1, 2-diamine)cobalt (III) sulphate is

(A) $[Co(en)_3]SO_4$ (B) $[Co(SO)_4(en)_3]$

(C) $[Co(en)_3](SO_4)_2$ (D) $[Co(en)_3]_2(SO_4)_3$ **CC0028**

Structural isomerism

29. Which of the following statement is **INCORRECT** regarding the following compound

$[Pt(NH_3)_4]$ $[PtCl_4]$

(A) It is the polymerisation isomer of $[Pt(NH_3)_3Cl_3]$

(B) E.A.N. of cationic part is equal to that of anionic part

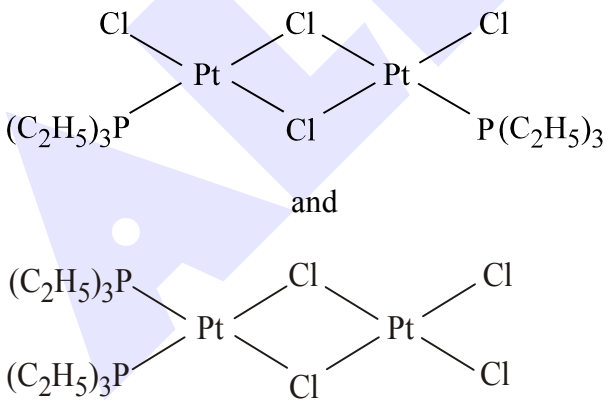
(C) It is the co-ordination isomer of $[Pt(NH_3)_3Cl]$ $[Pt(NH_3)Cl_3]$

(D) Synergic bonding is not involved in the complex **CC0029**

30. The type of isomerism present in pentaamminenitrochromium (III) chloride is :

(A) optical (B) linkage (C) hydrate (D) polymerisation **CC0030**

31. The complexes given below show:



(A) Optical isomerism (B) Co-ordination isomerism

(C) Geometrical isomerism (D) Co-ordination position isomerism **CC0031**

32. Which of the following complex shows ionization isomerism

(A) $[Cr(NH_3)_6]Cl_3$ (B) $[Cr(en)_2]Cl_2$

(C) $[Cr(en)_3]Cl_3$ (D) $[CoBr(NH_3)_5]SO_4$ **CC0032**

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33. Find the name of the hydrate isomer of $CrCl_3.6H_2O$, which is having lowest electrical conductivity excluding zero value of conductivity.

(A) Hexaaquachromium(III) chloride

(B) Tetraaquadichloridochromium(III) chloride dihydrate

(C) Pentaaquachloridochromium(III) chloride monohydrate

(D) Triaquatrichloridochromium(III) chloride trihydrate **CC0033**

Stereoisomerism

34. Which of the following complex shows optical isomerism -

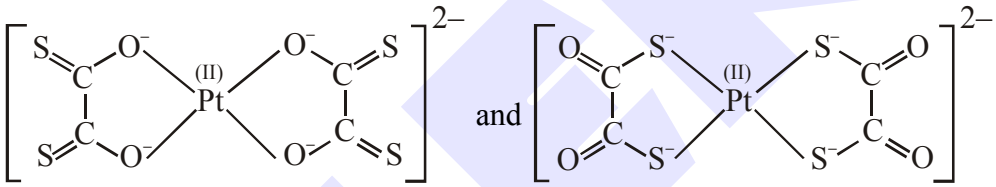
(A) $[Cd(CN)_4]^{2-}$ (B) $[Cr(H_2O)_3Cl_3].3H_2O$

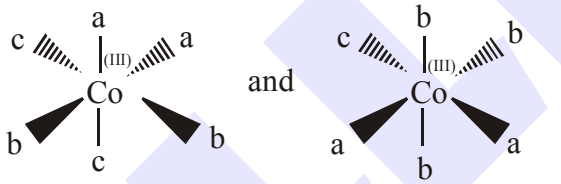
(C) $[Zn(gly)_2]^o$ (D) $[Ni(dmg)_2]^o$ **CC0034**

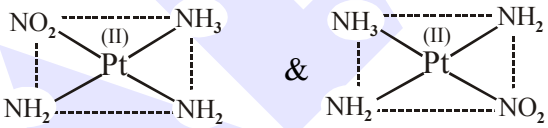
35. How many coordination isomers of $[Pt(NH_3)_4][PtCl_4]$ show geometrical isomerism.

(A) All (B) One (C) Two (D) None **CC0035**

36. Identify the pair of complex which are stereoisomer of each other -

(A) 

(B) 

(C) 

(D) All of the above **CC0036**

37. Find complex which have maximum number of stereoisomers -

(A) $[Ma_3b_3]$ (B) $[Ma_3b_2c]$ (C) $[Ma_2b_2c_2]$ (D) $[M(AA)a_2b_2]$

CC0037

38. In which of the following pairs both the complexes show optical isomerism ?

(A) cis-$[Cr(C_2O_4)_2Cl_2]^{3-}$, trans-$[Co(NH_3)_4Cl_2]$

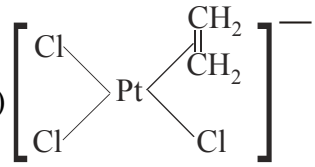
(B) $[Co(en)_3]Cl_3$, cis-$[Co(en)_2Cl_2]Cl$

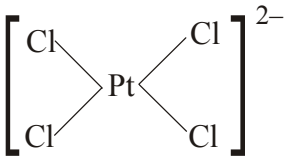
(C) $[PtCl(dien)]Cl$, $[NiCl_2Br_2]^{2-}$

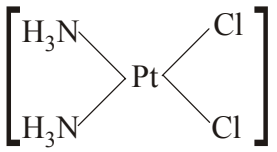
(D) $[Co(NO_3)_3(NH_3)_3]$, cis-$[Pt(en)_2Cl_2]$ **CC0038**

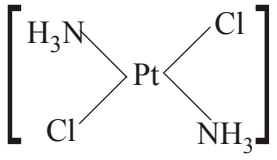
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39. Which of the following is considered to be an anticancer species ?

(A) 

(B) 

(C) 

(D)  **CC0039**

40. Which of the following can exhibit geometrical isomerism ?

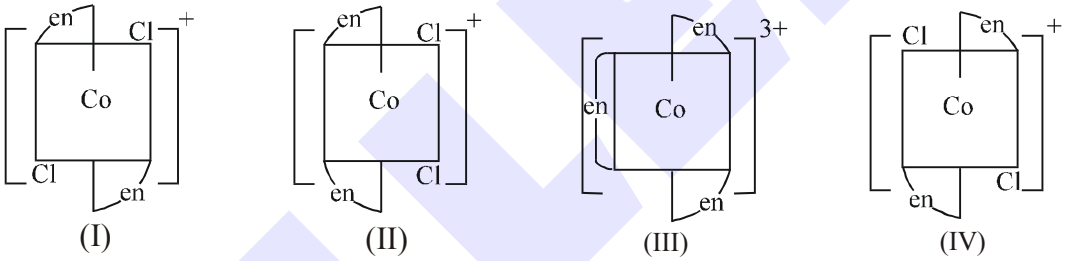
(A) $[MnBr_4]^{2-}$ (B) $[Pt(NH_3)_3Cl]^+$

(C) $[PtCl_2(P(C_2H_5)_3)_2]$ (D) $[Fe(H_2O)_5NOS]^{2+}$ **CC0040**

41. The oxidation state of Mo in its oxo-complex species $[Mo_2O_4(C_2H_4)_2(H_2O)_2]^{2-}$ is:

(A) +2 (B) +3 (C) +4 (D) +5 **CC0041**

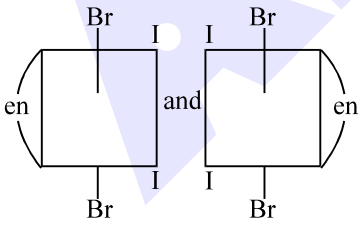
42. Which of the following ions are optically active?

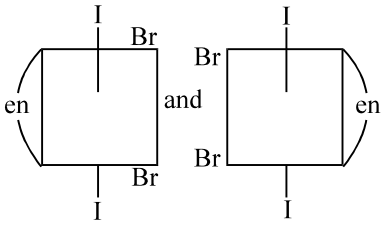


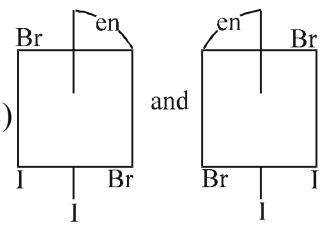
(I) (II) (III) (IV)

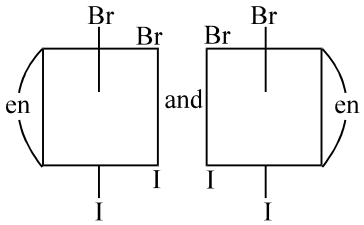
(A) I only (B) II only (C) II and III (D) IV only **CC0042**

43. The complex ion has two optical isomers. Their **CORRECT** configurations are:

(A) 

(B) 

(C) 

(D)  **CC0043**

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V.B.T/CFT

44. Which of the following complex is coloured and diamagnetic -

(A) MnO_4^{2-} (B) $[Ni(H_2O)_6]^{2+}$ (C) $[CrCl_6]^{3-}$ (D) CrO_4^{2-} **CC0044**

45. What is the hybridisation of Fe in $[Fe(CO)_4]^{-2}$

(A) d^2sp^3 (B) sp^3d^2 (C) sp^3d (D) sp^3 **CC0045**

46. One unknown complex has the spin only magnetic moment is of 1.73 BM. As per the C. F. T., complexe is.

(A) d^7, Oh-field, with Srong Feild Legand (B) d^9, sq.planar-field, with Srong Feild Legand

(C) d^9, Td,field with Weak Feild Legand (D) All of these **CC0046**

47. Identify tetrahedral species which has maximum magnetic moment value :

(A) $[CuCl_4]^{2-}$ (B) $[CoCl_4]^{2-}$ (C) $[FeCl_4]^{2-}$ (D) $[AlCl_4]^{-}$ **CC0047**

48. In the co-ordination compound $Na_4[Fe(CN)_5NOS]$ oxidation state of Fe is -

(A) +1 (B) +2 (C) +3 (D) +4 **CC0048**

49. Which one of the following complex is an outer orbital complex -

(A) $[Ni(H_2O)_6]^{+2}$ (B) $[Fe(H_2O)_5NO]SO_4$

(C) $[Fe(NH_3)_6]^{+2}$ (D) All of these **CC0049**

50. Which of the following is diamagnetic and sp^3 hybridised -

(A) $[NiCl_4]^{2-}$ (B) $[Ni(CN)_4]^{4-}$ (C) $[Ni(CN)_4]^{2-}$ (D) $[NiCl_2(PPh_3)_2]$

CC0050

51. $[Cr(H_2O)_6]Cl_3$ (atomic number of Cr = 24) has a magnetic moment of 3.83 B.M. The **CORRECT** distribution of 3d electrons in the chromium present in the complex is:

(A) $3d^1_{xy}, 3d^1_{yz}, 3d^1_{zx}$ (B) $3d^1_{xy}, 3d^1_{yz}, 3d^1_{z^2}$

(C) $3d^1_{(x^2-y^2)}, 3d^1_{z^2}, 3d^1_{xz}$ (D) $3d^1_{xy}, 3d^1_{(x^2-y^2)}, 3d^1_{yz}$ **CC0051**

52. $[Fe(H_2O)_6]^{+2}$ has Crystal Field Splitting Energy value 10,400 cm^{-1} and pairing energy value 17,600 cm^{-1} then it is :

(A) Low spin complex (B) Paramagnetic in nature

(C) Diamagnetic in nature (D) None of these **CC0052**

53. In which of the following coordination entities, the magnitude of Δ_0 [CFSE in octahedral field] will be maximum? :

(A) $[Co(CN)_6]^{3-}$ (B) $[Co(C_2O_4)_3]^{3-}$

(C) $[Co(H_2O)_6]^{3+}$ (D) $[Co(NH_3)_6]^{3+}$ **CC0053**

54. The number of unpaired electrons calculated in $[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ are :

(A) 4 and 4 (B) 0 and 2 (C) 2 and 4 (D) 0 and 4 **CC0054**

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55. An ion M^{2+}, forms the complexes $[M(H_2O)_6]^{2+}$, $[M(en)_3]^{2+}$ and $[MBr_6]^{4-}$, match the complex with the appropriate colour.

(A) Green, blue and red (B) Blue, red and green

(C) Green, red and blue (D) Red, blue and green **CC0055**

56. Formula of ferrocene is:

(A) $[Fe(CN)_6]^{4-}$ (B) $[Fe(CN)_6]^{3+}$ (C) $[Fe(CO)_5]$ (D) $[Fe(C_5H_5)_2]$

CC0056

57. $Ni(CO)_4$ and $[Ni(NH_3)_4]^{2+}$ do not differ in

(A) magnetic moment (B) oxidation number of Ni

(C) geometry (D) EAN **CC0057**

58. A complex of certain metal has the magnetic moment of 4.91 BM whereas another complex of the same metal with same oxidation state has zero magnetic moment. The metal ion could be

(A) Co^{2+} (B) Mn^{2+} (C) Fe^{2+} (D) Fe^{3+} **CC0058**

59. The tetrahedral $[CoI_4]^{2-}$ and square planar $[PdBr_4]^{2-}$ complex ions are respectively

(A) low spin, high spin (B) high spin, low spin

(C) both low spin (D) both high spin **CC0059**

60. Which one of the following species does not represent cationic species of vanadium formed in aqueous solution

(A) VO_2^+ (B) VO^{2+} (C) $[V(H_2O)_6]^{3+}$ (D) VO_2^{2+} **CC0060**

61. On treatment of $[Ni(NH_3)_4]^{2+}$ with concentrated HCl, two compounds I and II having the same formula, $[NiCl_2(NH_3)_2]$ are obtained, I can be converted into II by boiling with dilute HCl. A solution of I reacts with oxalic acid to form $[Ni(C_2O_4)(NH_3)_2]$ wheras II does not react. Point out the correct statement of the following

(A) I cis, II trans; both tetrahedral (B) I cis, II trans; both square planar

(C) I trans, II cis; both tetrahedral (D) I trans, II cis; both square planar **CC0061**

62. Among the following, the compound that is both paramagnetic and coloured is

(A) $K_2Cr_2O_7$ (B) $(NH_4)_2[TiCl_6]$ (C) $VOSO_4$ (D) $K_3[Cu(CN)_4]$

CC0062

63. The magnetic moment of $[NiX_4]^{2-}$ ion is found to be zero. Then the metal of the complex ion is (X = monodentate anionic ligand).

(A) sp^3 hybridised (B) spd^2 hybridised (C) dsp^2 hybridised (D) d^2sp hybridised

CC0063

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64. For which of the following types of d^n configuration, the number of unpaired electrons in octahedral complexes remains same irrespective of the ligand field strength.

(A) d^3 (B) d^4 (C) d^5 (D) d^6 **CC0064**

65. Which of the follwoing electronic arrangement gives the highest value of the magnetic moment?

(A) d^6, strong field (B) d^7, high spin (C) d^4, weak field (D) d^2, strong field

CC0065

66. Select appropriate ligand for given complex

$[\overset{III}{Co}(.....)_6]^{\pm x}$; $\mu = 0$ BM

(A) $C_2O_4^{2-}$ (B) en (C) H_2O (D) F^- **CC0066**

67. According to C.F.T. , ligands are treated as -

(A) Point charges (B) Lewis acids (C) Proton donor (D) All of the above

CC0067

68. Which of the following is correct electronic configuration of 3d orbital in excited state of central metal ion, when $[Ti(H_2O)_6]^{3+}$ absorbed yellow-green light.

(A) 3d | | | | 1 | |

(B) t_{2g}^1, e_g^0

(C) t_{2g}^1, e_g^1

(D) t_{2g}^0, e_g^1

CC0068

69. If $\lambda_{absorbed}$ for d-d transition is in order $[Ti(X)_6]^{3+} > [Ti(Y)_6]^{3+} > [Ti(Z)_6]^{3+}$.

Select correct order of strength of ligands (X, Y, Z are monodentate ligand)-

(A) Z > Y > X (B) X > Y > Z

(C) Z > X > Y (D) Not predictable **CC0069**

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EXERCISE # O-2

MORE THAN ONE MAY BE CORRECT

1. Which of the following exhibit geometrical isomerism (M stands for a metal, and a and b are achiral ligands)?

(A) Ma_2b_2 (Sq. Pl.) (B) Ma_4b_2 (C) Ma_5b (D) Ma_6 **CC0070**

2. Which of the following statement(s) is (are) **CORRECT** ?

(A) The oxidation state of iron in sodium nitroprusside $Na_2[Fe(CN)_5(NO)]$ is +II.

(B) $[Ag(CN)_2]^-$ is linear in shape.

(C) In $[Fe(H_2O)_6]^{3+}$, Fe is d^2sp^3 hybridized.

(D) In $Ni(CO)_4$, the oxidation state of Ni is zero. **CC0071**

3. Which of the following compound(s) show(s) optical isomerism.

(A) $[Pt(bn)_2]^{2+}$ (B) $[CrCl_2(en)_2]^+$ (C) $[Co(en)_3]\,[CoF_6]$ (D) $[Zn(gly)_2]$ **CC0072**

4. Select **INCORRECT** statement(s) for $[Cu(CN)_4]^{3-}$, $[Cd(CN)_4]^{2-}$ and $[Cu(NH_3)_4]^{2+}$ complex ion.

(A) Both $[Cd(CN)_4]^{2-}$ and $[Cu(NH_3)_4]^{2+}$ have square planar geometry

(B) $[Cu(CN)_4]^{3-}$ and $[Cu(NH_3)_4]^{2+}$ have equal no. of unpaired electron

(C) $[Cu(CN)_4]^{3-}$ and $[Cd(CN)_4]^{2-}$ can be separated from the mixture on passing H_2S gas.

(D) All the three complexes have magnetic moment equal to zero. **CC0073**

5. Which of the following will have two stereoisomeric forms?

(A) $[Cr(NO_3)_3(NH_3)_3]$ (B) $K_3[Fe(C_2O_4)_3]$

(C) $[CoCl_2(en)_2]^+$ (D) $[CoBrCl(Ox)_2]^{3-}$ **CC0074**

6. Which is / are **NOT** correctly matched.

	Complex compounds	**IUPAC name**
(A)	$K[CrF_4O]$	Potassium tetrafluoridooxidochromate(V)
(B)	$Na[BH(OCH_3)_3]$	Sodium hydridotrimethoxyborate(III)
(C)	$[Be(CH_3–CO–CH_2–CO–C_6H_5)_2]°$	Bis(benzoylacetonato)beryllium(III)
(D)	$H[AuCl_4]$	Hydrogen tetrachloroaurate(III) **CC0075**

7. Which of the following statement(s) is/are **INCORRECT**

(A) In $[CoBrCl(en)_2]^+$ geometrical isomerism exists, while optical isomerism does not exist

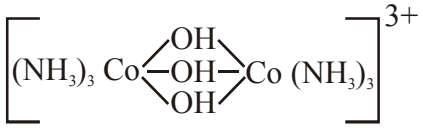
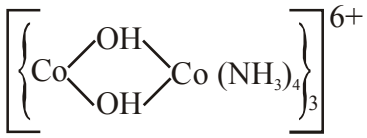
(B) Potassium aquadicyanidosuperoxidoperoxidochromate(III) is IUPAC name for $K_2[Cr(CN)_2O_2(O_2)(H_2O)]$

(C) There are 3 geometrical isomers and 15 stereoisomers possible for $[Pt(NO_2)(NH_3)(NH_2OH)(py)]^+$ and [PtBr C*l* I $(NO_2)(NH_3)(py)$] respectively

(D) cis and trans forms are not diastereomers of each other **CC0076**

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8. Which of the following complexes are polymerisation isomers :

(A)  and 

(B) $[Pt(NH_3)_4]\,[PtCl_4]$ and $[Pt(NH_3)_4]\,[Pt(NH_3)Cl_3]_2$

(C) $[Pt(NH_3)_2Cl_2]$ and $[Pt(NH_3)_3Cl]_2\,[PtCl_4]$

(D) $[Pt(NH_3)_2Cl_4]$ and $[Pt(NH_3)_2Cl_2]$ **CC0077**

9. Which of the following is **CORRECT** about

Tetraamminedithiocyanato-Scobalt(III) tris(oxalato)cobaltate(III)

(A) formula of the complex is $[Co(SCN)_2(NH_3)_4][Co(ox)_3]$

(B) It is a chelating complex and show linkage isomerism.

(C) It shows optical isomerism.

(D) It shows geometrical isomerism. **CC0078**

10. Which is **CORRECT** statement(s)?

(A) $[Ag(NH_3)_2]^+$ is linear with sp hybridised Ag^+ ion

(B) $NiCl_4^{2-}$, VO_4^{3-} and MnO_4^- have tetrahedral geometry

(C) $[Cu(NH_3)_4]^{2+}$, $[Pt(NH_3)_4]^{2+}$ & $[Ni(CN)_4]^{2-}$ have dsp^2 hybridisation of the metal ion

(D) $Fe(CO)_5$ has trigonal bipyramidal structure with dz^2sp^3 hybridised iron. **CC0079**

11. In which of the following complex(s) spin only magnetic moment is independent , from the nature of ligand. (L = monodented ligand) -

(A) $[\overset{II}{Ni}L_4]$ (B) $[\overset{II}{Ni}L_6]$ (C) $[\overset{III}{Fe}L_6]$ (D) $[\overset{III}{Cr}L_6]$ **CC0080**

12. Which of the following compound(s) can show optical isomerism ?

(A) $[PtCl_4]^{2-}$ (B) $[PtCl_2(NH_3)_2]$ (C) $[Fe(EDTA)]^-$ (D) $[Fe(en)_3]^{3+}$

CC0081

13. Which of the following compounds are resolvable into d or ℓ-forms?

(A) $[ZnCl_2(en)]$ (B) $[Be(acac)_2]$ (C) $[Co(gly)_3]$ (D) $[Cr(C_2O_4)_3]^{3-}$

CC0082

14. Which of the following ion is/are diamagnetic and non planar ?

(A) $[Ni(CN)_4]^{2-}$ (B) MnO_4^- (C) $[Cu(NH_3)_4]^{2+}$ (D) CrO_4^{2-} **CC0083**

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EXERCISE # S-1

INTEGER TYPE

1. Find number of ligands which is / are stronger ligand as compared to NH_3

NO_2^-, H_2O, NO_3^-, F^-, $C_2O_4^{2-}$, en, Cl^-, $\overline{C}N$ **CC0084**

2. If crystal field stablization energy of $[ML_6]^{+n}$ is $-0.8\ \Delta_O$.

Find minimum number of electrons in t_{2g} orbitals of metal ion ? **CC0085**

3. Find number of Co–N linkage in,

Pentaamminecobalt(III)–μ–amidodiamminetriaquacobalt(III) chloride. **CC0086**

4. Find the EAN value of central atom of $[Fe(\pi\text{–}C_4H_4)\ (CO)_3]$ **CC0087**

5. Find the maximum number of atoms lying in one plane for $[Cr(CN)_6]^{3-}$ **CC0088**

6. Select complex in which metal–carbon linkage(s) is / are present :

$\left[\overset{III}{Fe}(EDTA)\right]^-$, $K[PtCl_3(\pi\text{–}C_2H_4)]$, $[Cr(acac)_3]^o$, $[Fe(CO)_5]$, $[Co(C_2H_5O)_6]^{3-}$, $[Ni(dmg)_2]$

CC0089

7. Find out the total number of geometrical isomers of $[Co(H_2O)_3Cl_3]$. **CC0090**

8. Find the value of E.A.N of $[Pd(CO)_4]^{+2}$ (atomic number = 46) : **CC0091**

9. A co-ordination compound have magnetic moment 5.92 B.M. Find out the number of unpaired electron(s) in the compound. **CC0092**

10. Find the number of optically active isomers for $[Pd(en)_2(NH_3)(H_2O)]^{4+}$ cation. **CC0093**

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EXERCISE # S-2

MATCH THE COLUMN :

1. Match the complexes in column-I with the EAN of central atom in column-II:

	Column-I		**Column-II**
(A)	$[Fe(CO)_4]^{2-}$	(P)	34
(B)	$[Co(NH_3)_5Cl]Cl_2$	(Q)	35
(C)	$K_2[Ni(CN)_4]$	(R)	36
(D)	$[Cu(NH_3)_4]^{2+}$	(S)	37 **CC0094**

2.

	Column - I		**Column -II**
(A)	Na_2 $[Fe(CN)_5$ NO]	(P)	$\mu = 0$ B.M.
(B)	$[Fe(H_2O)_5$ NO] SO_4	(Q)	octahedral
(C)	$[Ag(CN)_2]^-$	(R)	$\mu = \sqrt{15}$ B.M.
(D)	$K_4[Fe(CN)_6]$	(S)	NO^+ ligand **CC0095**

3. Match the complexes in column I with their stereoproperties is column II

	Column I		**Column II**
(A)	$[CoCl_3(NH_3)_3]$	(P)	Has a facial isomer
(B)	$[Cr(ox)_3]^{3-}$	(Q)	Cis form is optically active
(C)	$[CrCl_2(ox)_2]$	(R)	Trans form is optically inactive
(D)	$[RhCl_3(Py)_3]$	(S)	Has a meridional form
		(T)	Two optically active isomer **CC0096**

4. Match each coordination compound in List-I with co-ordination number of central metal/ion from List-II and select the correct answer using the code given below the lists.

List-I	**List-II**
(P) $[Co(en)_3]^{3+}$	(1) 6
(Q) $[Ca(EDTA)]^{2-}$	(2) 4
(R) $[Ni(CO)_4]$	(3) 2
(S) $[Ag(NH_3)_2]Cl$	(4) 5 **CC0097**

Code :

	P	**Q**	**R**	**S**
(A)	2	1	2	3
(B)	1	1	2	3
(C)	1	4	2	3
(D)	1	1	3	2

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5. Match the List-I with List-II :

List-I	List-II
(P) Ferrocene	(1) Iron present
(Q) $Mn_2(CO)_{10}$	(2) Cobalt
(R) Vitamine B_{12}	(3) Metal-Metal bonding
(S) Haemoglobin	(4) Sandwich structure

CC0098

Code :

	P	Q	R	S
(A)	4	3	1	2
(B)	1	3	1	2
(C)	1,4	3	2	1
(D)	1	3	4	2

6. Match the List-I with List-II :

List-I	List-II
(P) $EDTA^{4-}$	(1) N-donor atom
(Q) en	(2) Chelate ligand with same donor site
(R) gly^-	(3) Bidentate with different donor atom
(S) amide	(4) Hexadentate

CC0099

Code :

	P	Q	R	S
(A)	4	2	1	3
(B)	2	1	3	4
(C)	4	1	2	3
(D)	4	2	3	1

Assertion Reason :

7. **Statement-1**: Complexes containing three bidentate groups such as $[Cr(ox)_3]^{3-}$ and $[Co(en)_3]^{3+}$ do not show optical activity.

Statement-2: Octahedral complex, $[Co(NH_3)_4Cl_2]Cl$ shows geometrical isomerism.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **CC0100**

8. **Statement-1**: After splitting of d-orbitals during complex formation, the orbitals form two sets of orbitals t_{2g} and e_g in an octahedral field.

Statement-2: Splitting of d-orbitals occur only in the case of strong field ligands such as CN^-.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **CC0101**

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9. **Statement-1**: $[Ti(H_2O)_6]^{3+}$ is coloured while $[Sc(H_2O)_6]^{3+}$ is colourless.

Statement-2: d–d transition is not possible in $[Sc(H_2O)_6]^{3+}$ because no d-electron is present while possible for Ti^{3+} having d^1 system.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **CC0102**

Comprehension (10 to 12)

Ligands are neutral or ionic species capable of donating at least one electron pair to central metal. Hence ligands can be of different denticities.

10. For a given metal M^{3+} coordination number is six, then for which set of ligands, complex will be most stable-

(A) $6H_2O$ (B) $6F^-$ (C) $EDTA^{4-}$ (D) $2H_2O$ and $2C_2O_4^{2-}$ **CC0103**

11. $[Mn(CO)_5]$ can attain more stability by :

(A) Oxidation of itself (B) Reduction of itself

(C) Dimerization (D) Both (B) and (C) **CC0104**

12. The metal cation that has least tendency to accept electron pair from NH_3 is

(A) Fe^{3+} (B) Rh^{3+} (C) Zn^{2+} (D) Ba^{++} **CC0105**

Comprehension (13 to 15)

Complex compounds are molecular compounds which retain their identities even when dissolved in water. They do not give all the simple ions in solution but instead furnish complex ions with complicated structures. The complex compounds are often called coordination compounds because certain groups called ligands are attached to the central metal ion by coordinate or dative bonds. Coordination compounds exhibit isomerism, both structural and stereoisomerism. The structure, magnetic property, colour and electrical properties of complexes are explained by various theories.

13. Arrange the following compounds in order of their Molar conductance:

(I) $K[Co(NO_2)_4(NH_3)_2]$ (II) $[Cr(ONO)_3(NH_3)_3]$

(III) $[Cr(NO_2)(NH_3)_5]_3[Co(NO_2)_6]_2$ (IV) $Mg[Cr(NO_2)_5(NH_3)]$

(A) II < I < IV < III (B) I < II < III < IV

(C) II < I < III < IV (D) IV < III < II < I **CC0106**

14. The oxidation number and coordination number of chromium in the following complex is $[Cr(C_2O_4)_2(NH_3)_2]^{1-}$

(A) O.N. = + 4, C.N. = 4 (B) O.N. = +3, C.N. = 4

(C) O.N. = – 1, C.N. = 4 (D) O.N. = + 3, C.N. = 6 **CC0107**

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15. In which of the following pairs, both the complexes have the same geometry but different hybridisation

(A) $[NiCl_4]^{2-}$, $[Ni(CN)_4]^{2-}$ (B) $[CoF_6]^{3-}$, $[Co(NH_3)_6]^{3+}$

(C) $[Ni(CO)_4]$, $[Ni(CN)_4]^{2-}$ (D) $[Cu(NH_3)_4]^{2+}$, $[Ni(NH_3)_6]^{2+}$ **CC0108**

Matching list type 1 × 3Q. (Three list type Question)

The following column I, II, III represent the different type of observations based on CFT in complex compounds.

Answer the questions that follow

Column-I - Crystal field stablization energy (CFSE) (neglecting PE in all cases)

Column-II - Electronic configuration

Column-III - Type of complex

Column - I CFSE (neglecting PE in all cases	**Column - II** Electronic Configuration	**Column - III** Type of Complex
(I) $-0.4\ \Delta_O$	(i) t_{2g}^5, e_g^0	(P) High spin & Paramagnetic
(II) $-2.0\ \Delta_O$	(ii) t_{2g}^4, e_g^0	(Q) Low spin & Paramagnetic
(III) $-2.4\ \Delta_O$	(iii) t_{2g}^6, e_g^0	(R) High spin & Diamagnetic
(IV) $-1.2\ \Delta_O$	(iv) t_{2g}^4, e_g^2	(S) Low spin & Diamagnetic

16. For sodium nitroprusside complex the only **CORRECT** combination is

(A) (III), (iv), (Q) (B) (III), (iii), (S) (C) (III), (iii), (R) (D) (II), (iii), (Q)

CC0109

17. For $[Co(H_2O)_3F_3]$ complex the only **CORRECT** combination is.

(A) (I), (iv), (Q) (B) (II), (iv), (S) (C) (III), (ii), (R) (D) (I), (iv), (P)

CC0110

18. For $[Mn(CN)_6]^{4-}$ complex the only **CORRECT** combination is.

(A) (IV), (i), (S) (B) (II), (i), (R) (C) (I), (i), (S) (D) (II), (i), (Q)

CC0111

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EXERCISE # JEE-MAIN

1. Which among the following will be named as dibromidobis (ethylene diamine) chromium (III) bromide ? **[AIEEE–2012]**

(1) $[Cr(en)Br_2]Br$ (2) $[Cr(en)_3]Br_3$ (3) $[Cr(en)_2Br_2]Br$ (4) $[Cr(en)Br_4]^-$

CC0140

2. The complex ion

$[Pt(NO_2)(Py)(NH_3)(NH_2OH)]^+$ will give :- **[J-MAIN–2012, Online]**

(1) 4 isomers (Geometrical) (2) 2 isomers (Geometrical)

(3) 3 isomers (Geometrical) (4) 6 isomers (Geometrical) **CC0141**

3. Which of the following complex ions will exhibit optical isomerism? **[J-MAIN–2012, Online]**

(en = 1, 2-diamine ethane)

(1) $[Co(en)_2Cl_2]^+$ (2) $[Zn(en)_2]^{2+}$ (3) $[Co(NH_3)_4Cl_2]^+$ (4) $[Cr(NH_3)_2Cl_2]^+$

CC0142

4. Which of the following complex species is **NOT** expected to exhibit optical isomerism ?**[J-MAIN–2013]**

(1) $[Co(en)_3]^{3+}$ (2) $[Co(en)_2Cl_2]^+$ (3) $[Co(NH_3)_3Cl_3]$ (4) $[Co(en)(NH_3)_2Cl_2]^+$

CC0143

5. Type of isomerism which exists between $[Pd(C_6H_5)_2(SCN)_2]$ and $[Pd(C_6H_5)_2(NCS)_2]$ is : **[J-MAIN–2013, Online]**

(1) Solvate isomerism (2) Ionisation isomerism

(3) Linkage isomerism (4) Coordination isomerism **CC0144**

6. Which of the following is diamagnetic ? **[J-MAIN–2013, Online]**

(1) $[CoF_6]^{3-}$ (2) $[FeF_6]^{3-}$ (3) $[Fe(CN)_6]^{3-}$ (4) $[Co(Ox)_3]^{3-}$

CC0145

7. The magnetic moment of the complex anion $[Cr^{I}(NO)(NH_3)(CN)_4]^{2-}$ is: **[J-MAIN–2013, Online]**

(1) 2.82 BM (2) 5.91 BM (3) 1.73 BM (4) 3.87 BM **CC0146**

8. The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1, L_2, L_3 and L_4 absorb wavelength in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is: **[J-MAIN–2014]**

(1) $L_3 < L_2 < L_4 < L_1$ (2) $L_1 < L_2 < L_4 < L_3$ (3) $L_4 < L_3 < L_2 < L_1$ (4) $L_1 < L_3 < L_2 < L_4$

CC0147

9. The equation which is balanced and represents the **CORRECT** product (s) is : **[J-MAIN–2014]**

(1) $[Mg(H_2O)_6]^{2+} + (EDTA)^{4-} \xrightarrow{\text{excess NaOH}} [Mg(EDTA)]^{2+} + 6H_2O$

(2) $CuSO_4 + 4KCN \rightarrow K_2[Cu(CN)_4] + K_2SO_4$

(3) $Li_2O + 2KCl \rightarrow 2LiCl + K_2O$

(4) $[CoCl(NH_3)_5]^+ + 5H^+ \rightarrow Co^{2+} + 5NH_4^+ + Cl^-$ **CC0148**

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10. The **CORRECT** statement about the magnetic properties of $[Fe(CN)_6]^{3-}$ and $[FeF_6]^{3-}$ is : (Z = 26).
(1) $[Fe(CN)_6]^{3-}$ is paramagnetic, $[FeF_6]^{3-}$ is diamagnetic. **[J-MAIN–2014, Online]**
(2) both are diamagnetic.
(3) $[Fe(CN)_6]^{3-}$ is diamagnetic, $[FeF_6]^{3-}$ is paramagnetic.
(4) both are paramagnetic **CC0149**

11. An octahedral complex of Co^{3+} is diamagnetic. The hybridisation involved in the formation of the complex is : **[J-MAIN–2014]**
(1) d^2sp^3 (2) dsp^3d (3) dsp^2 (4) sp^3d^2 **CC0150**

12. Which of the following name formula combinations is **NOT CORRECT**? **[J-MAIN–2014, Online]**

Formula	Name
(1) $K[Cr(NH_3)_2Cl_4]$	Potassium diammine Tetrachlorochromate (III)
(2) $[Co(NH_3)_4(H_2O)I]SO_4$	Tetraammine aquaiodo cobalt (III) sulphate
(3) $[Mn(CN)_5]^{2-}$	Pentacyanomagnate (II) ion
(4) $K_2[Pt(CN)_4]$	Potassium tetracyanoplatinate(II) **CC0151**

13. Consider the coordination compound, $[Co(NH_3)_6]Cl_3$. In the formation of this complex, the species which acts as the Lewis acid is : **[J-MAIN–2014, Online]**
(1) $[Co(NH_3)_6]^{3+}$ (2) NH_3 (3) Co^{3+} (4) $C\ell^-$ **CC0152**

14. Among the following species the one which causes the highest CFSE, Δ_0 as a ligand is :- **[J-MAIN–2014, Online]**
(1) CN^- (2) NH_3 (3) CO (4) F^- **CC0153**

15. Which one of the following complexes will most likely absorb visible light ? **[J-MAIN–2014, Online]**
(At. nos. Sc = 21, Ti = 22, V = 23, Zn = 30) :-
(1) $[Ti(NH_3)_6]^{4+}$ (2) $[V(NH_3)_6]^{3+}$ (3) $[Zn(NH_3)_6]^{2+}$ (4) $[Sc(H_2O)_6]^{3+}$
CC0154

16. Nickel(Z = 28) combines with a uninegative monodentate ligand to form a diamagnetic complex $[NiL_4]^{2-}$. The hybridisation involved and the number of unpaired electrons present in the complex are respectively : **[J-MAIN–2014, Online]**
(1) sp^3, zero (2) sp^3. two (3) dsp^2 one (4) dsp^2, zero **CC0155**

17. The number of geometrical isomers that can exist for square planar [Pt (Cl) (py) (NH_3) $(NH_2OH)]^+$ is (py = pyridine) : **[J-MAIN–2015]**
(1) 4 (2) 6 (3) 2 (4) 3 **CC0156**

18. The color of $KMnO_4$ is due to : **[J-MAIN–2015]**
(1) $L \rightarrow M$ charge transfer transition (2) $\sigma - \sigma^*$ transition
(3) $M \rightarrow L$ charge transfer transition (4) d – d transition **CC0157**

19. Which of the following complex ions has electrons that are symmetrically filled in both t_{2g} and e_g orbitals ? **[J-MAIN–2015, Online]**
(1) $[CoF_6]^{3-}$ (2) $[Mn(CN)_6]^{4-}$ (3) $[FeF_6]^{3-}$ (4) $[Co(NH_3)_6]^{2+}$
CC0158

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20. When concentrated HCl is added to an aqueous solution of $CoCl_2$, its colour changes from reddish pink to deep blue. Which complex ion gives blue colour in this reaction ? **[J-MAIN–2015, Online]**

(1) $[Co(H_2O)_6]^{2+}$ (2) $[CoCl_6]^{3-}$ (3) $[CoCl_4]^{2-}$ (4) $[CoCl_6]^{4-}$ **CC0159**

21. The **CORRECT** statement on the isomerism associated with the following complex ions,

(a) $[Ni(H_2O)_5NH_3]^{2+}$ (b) $[Ni(H_2O)_4(NH_3)_2]^{2+}$ and (c) $[Ni(H_2O)_3(NH_3)_3]^{2+}$ is :

(1) (a) and (b) show geometrical and optical isomerism

(2) (b) and (c) show geometrical and optical isomerism

(3) (a) and (b) show only geometrical Isomerism **[J-MAIN–2015, Online]**

(4) (b) and (c) show only geometrical Isomerism **CC0160**

22. Which one of the following complexes shows optical isomerism :- **[J-MAIN–2016]**

(1) $[Co(NH_3)_4Cl_2]Cl$ (2) $[Co(NH_3)_3Cl_3]$ (3) cis$[Co(en)_2Cl_2]Cl$ (4) trans$[Co(en)_2Cl_2]Cl$

(en = ethylenediamine) **CC0161**

23. The pair having the same magnetic moment is:- **[J-MAIN–2016]**

[At. No.: Cr = 24, Mn = 25, Fe = 26, Co = 27]

(1) $[CoCl_4]^{2-}$ and $[Fe(H_2O)_6]^{2+}$ (2) $[Cr(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$

(3) $[Cr(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{2+}$ (4) $[Mn(H_2O)_6]^{2+}$ and $[Cr(H_2O)_6]^{2+}$ **CC0162**

24. Which one of the following complexes will consume more equivalents of aqueous solution of $Ag(NO_3)$? **[J-MAIN–2016, Online]**

(1) $[Cr(H_2O)_6]Cl_3$ (2) $Na_2[CrCl_5(H_2O)]$ (3) $Na_3[CrCl_6]$ (4) $[Cr(H_2O)_5Cl]Cl_2$

CC0163

25. Identify the **CORRECT** trend given below : **[J-MAIN–2016, Online]**

(Atomic No.= Ti : 22, Cr : 24 and Mo : 42)

(1) Δ_0 of $[Cr(H_2O)_6]^{2+} < [Mo(H_2O)_6]^{2+}$ and Δ_0 of $[Ti(H_2O)_6]^{3+} < [Ti(H_2O)_6]^{2+}$

(2) Δ_0 of $[Cr(H_2O)_6]^{2+} < [Mo(H_2O)_6]^{2+}$ and Δ_0 of $[Ti(H_2O)_6]^{3+} > [Ti(H_2O)_6]^{2+}$

(3) Δ_0 of $[Cr(H_2O)_6]^{2+} > [Mo(H_2O)_6]^{2+}$ and Δ_0 of $[Ti(H_2O)_6]^{3+} > [Ti(H_2O)_6]^{2+}$

(4) Δ_0 of $[Cr(H_2O)_6]^{2+} > [Mo(H_2O)_6]^{2+}$ and Δ_0 of $[Ti(H_2O)_6]^{3+} < [Ti(H_2O)_6]^{2+}$

CC0164

26. $[Co_2(CO)_8]$ displays:- **[J-MAIN–2017, Online]**

(1) no Co-Co bond, four terminal CO and four bridging CO

(2) one Co-Co bond, six terminal CO and two bridging CO

(3) no Co–Co bond, six terminal CO and two bridging CO

(4) one Co-Co bond, four terminal CO and four bridging CO **CC0165**

27. On treatment of 100 mL of 0.1 M solution of $CoCl_3 . 6H_2O$ with excess $AgNO_3$; 1.2×10^{22} ions are precipitated. The complex is :- **[J-MAIN–2017, Offline]**

(1) $[Co(H_2O)_4 Cl_2]Cl.2H_2O$ (2) $[Co(H_2O)_3Cl_3].3H_2O$

(3) $[Co(H_2O)_6]Cl_3$ (4) $[Co(H_2O)_5Cl]Cl_2.H_2O$ **CC0166**

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28. The pair of compounds having metal in their highest oxidation state is : **[J-MAIN–2017, Online]**

(1) $[NiCl_4]^{2-}$ and $[CoCl_4]^{2-}$ (2) $[Fe(CN)_6]^{3-}$ and $[Cu(CN)_4]^{2-}$

(3) $[FeCl_4]^{-}$ and Co_2O_3 (4) MnO_2 and CrO_2Cl_2 **CC0167**

29. The oxidation states of Cr in $[Cr(H_2O)_6]Cl_3$,$[Cr(C_6H_6)_2]$, and $K_2[Cr(CN)_2(O)_2(O_2)(NH_3)]$ respectively are : **[J-MAIN–2018, Offline]**

(1) +3, +2, and +4 (2) +3, 0, and +6 (3) +3, 0, and +4 (4) +3, +4, and +6

CC0168

30. Consider the following reaction and statements : **[J-MAIN–2018, Offline]**

$[Co(NH_3)_4Br_2]^+ + Br^- \rightarrow [Co(NH_3)_3Br_3] + NH_3$

(I) Two isomers are produced if the reactant complex ion is a *cis*-isomer.

(II) Two isomers are produced if the reactant complex ion is a *trans*-isomer.

(III) Only one isomer is produced if the reactant complex ion is a *trans*-isomer.

(IV) Only one isomer is produced if the reactant complex ion is a *cis*-isomer.

The correct statements are :

(1) (I) and (III) (2) (III) and (IV) (3) (II) and (IV) (4) (I) and (II)

CC0169

31. For 1 molal aqueous solution of the following compounds, which one will show the highest freezing point ? **[J-MAIN–2018, Offline]**

(1) $[Co(H_2O)_5Cl]Cl_2.H_2O$ (2) $[Co(H_2O)_4Cl_2]Cl.2H_2O$

(3) $[Co(H_2O)_3Cl_3].3H_2O$ (4) $[Co(H_2O)_6]Cl_3$ **CC0170**

32. The total number of possible isomers for square-planar $[Pt(Cl)(NO_2)(NO_3)(SCN)]^{2-}$ is :- **[J-MAIN–2018, Online]**

(1) 16 (2) 8 (3) 24 (4) 12 **CC0171**

33. The correct order of spin-only magnetic moments among the following is : **[J-MAIN–2018, Online]**

(Atomic number : Mn = 25, Co = 27, Ni= 28, Zn = 30)

(1) $[ZnCl_4]^{2-} > [NiCl_4]^{2-} > [CoCl_4]^{2-} > [MnCl_4]^{2-}$

(2) $[CoCl_4]^{2-} > [MnCl_4]^{2-} > [NiCl_4]^{2-} > [ZnCl_4]^{2-}$

(3) $[MnCl_4]^{2-} > [CoCl_4]^{2-} > [NiCl_4]^{2-} > [ZnCl_4]^{2-}$

(4) $[NiCl_4]^{2-} > [CoCl_4]^{2-} > [MnCl_4]^{2-} > [ZnCl_4]^{2-}$ **CC0172**

34. The correct combination is **[J-MAIN–2018, Online]**

(1) $[Ni(CN)_4]^{2-}$ – tetrahedral ;
$[Ni(CO)_4]$ – paramagnetic

(2) $[NiCl_4]^{2-}$ – paramagnetic ;
$[Ni(CO)_4]$ – tetrahedral

(3) $[NiCl_4]^{2-}$ – diamagnetic ;
$[Ni(CO)_4]$ – square-planar

(4) $[NiCl_4]^{2-}$ – square-planar ;
$[Ni(CN)_4]^{2-}$ – paramagnetic **CC0173**

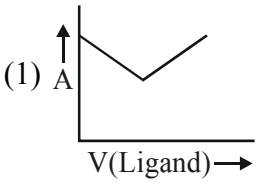
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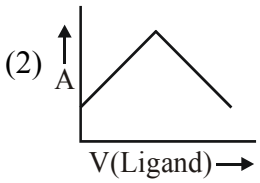
35. Which of the following complexes will show geometrical isomerism ? **[J-MAIN–2018, Online]**

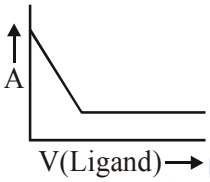
(1) Potassium amminetrichloroplatinate(II)
(2) Aquachlorobis (ethylenediamine) cobalt(II) chloride
(3) Potassium tris(oxalato) chromate(III)
(4) Pentaaquachlorochromium(III) chloride

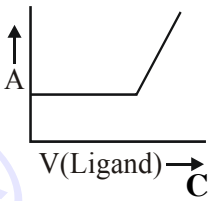
CC0174

36. In a complexometric titration of metal ion with ligand M(Metal ion) + L(Ligand) → C(Complex) end point is estimated spectrophotometrically (through light absorption). If 'M' and 'C' do not absorb light and only 'L' absorbs, then the titration plot between absorbed light (A) versus volume of ligand 'L' (v)would look like :- **[J-MAIN–2018, Online]**

(1) 

(2) 

(3) 

(4) 

CC0175

37. In Wilkinson's catalyst, the hybridization of central metal ion and its shape are respectively

(1) dsp^2, square planar
(2) sp^3d, trigonal bipyramidal
(3) sp^3, tetrahedral
(4) d^2sp^3, octahedral **[J-MAIN–2018, Online]**

CC0176

38. The molecule in which hybrid MOs involve only one d-orbital of the central atom is :-

[JEE-MAINS-2020]

(1) $[Ni(CN)_4]^{2-}$ (2) $[CrF_6]^{3-}$ (3) BrF_5 (4) XeF_4

CC0229

39. The theory that can completely/properly explain the nature of bonding in $[Ni(CO)_4]$ is:

(1) Werner's theory
(2) Crystal field theory **[JEE-MAINS-2020]**
(3) Valence bond theory
(4) Molecular orbital theory

CC0232

40. The volume (in mL) of 0.125 M $AgNO_3$ required to quantitatively precipitate chloride ions in 0.3 g of $[Co(NH_3)_6]Cl_3$ is _____. **[JEE-MAINS-2020]**

$^M[Co(NH_3)_6]Cl_3 = 267.46$ g/mol

$^MAgNO_3 = 169.87$ g/mol **CC0233**

41. Considering that $\Delta_0 > P$, the magnetic moment (in BM) of $[Ru(H_2O)_6]^{2+}$ would be_____ .

[JEE-MAINS-2020]

CC0234

42. The calculated magnetic moments (spin only value) for species $[FeCl_4]^{2-}$, $[Co(C_2O_4)_3]^{3-}$ and MnO_4^{2-} respectively are : **[JEE-MAINS-2021]**

(1) 5.82, 0 and 0 BM
(2) 4.90, 0 and 1.73 BM
(3) 5.92, 4.90 and 0 BM
(4) 4.90, 0 and 2.83 BM

CC0235

43. Which one of the following metal complexes is most stable? **[JEE-MAINS-2021]**

(1) $[Co(en)(NH_3)_4]Cl_2$ (2) $[Co(en)_3]Cl_2$ (3) $[Co(en)_2(NH_3)_2]Cl_2$ (4) $[Co(NH_3)_6]Cl_2$

CC0236

44. The number of geometrical isomers possible in triamminetrinitrocobalt (III) is X and in trioxalatochromate (III) is Y. Then the value of X + Y is _____. **[JEE-MAINS-2021]**

CC0237

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45. Arrange the following Cobalt complexes in the order of increasing Crystal Field Stabilization Energy (CFSE) value. **[JEE-MAINS-2021]**

Complexes : $[\underset{A}{CoF_6}]^{3-}$,$[\underset{B}{Co(H_2O)_6}]^{2+}$,$[\underset{C}{Co(NH_3)_6}]^{3+}$ and $[Co(en)_3]^{3+}$

Choose the correct option :

(1) A < B < C < D (2) B < A < C < D (3) B < C < D < A (4) C < D < B < A **CC0238**

46. Which one of the following complexes is violet in colour? **[JEE-MAINS-2021]**

(1) $[Fe(CN)_6]^{4-}$ (2) $[Fe(SCN)_6]^{4-}$ (3) $Fe_4[Fe(CN_6)]_3.H_2O$ (4) $[Fe(CN)_5NOS]^{4-}$

CC0239

47. Octahedral complexes of copper (II) undergo structural distortion (Jahn-Teller). Which one of the given copper (II) complexes will show the maximum structural distortion ?

(en-ethylenediamine; $H_2N\text{-}CH_2\text{-}CH_2\text{-}NH_2$) **[J-Main 2022]**

(A) $[Cu(H_2O)_6]SO_4$ (B) $[Cu(en)(H_2O)_4]SO_4$

(C) cis-$[Cu(en)_2Cl_2]$ (D) trans-$[Cu(en)_2Cl_2]$ **CC0243**

48. Match List I with List II. **[J-Main 2022]**

	List I (Complex)		**List II (Hybridization)**
A	$Ni(CO)_4$	I	sp^3
B	$[Ni(CN)_4]^{2-}$	II	sp^3d^2
C	$[Co(CN)_6]^{3-}$	III	d^2sp^3
D	$[CoF_6]^{3-}$	IV	dsp^2

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-III, D-II (B) A-I, B-IV, C-III, D-II

(C) A-I, B-IV, C-II, D-III (D) A-IV, B-I, C-II, D-III

CC0244

49. Total number of relatively more stable isomer(s) possible for octahedral complex $[Cu(en)_2(SCN)_2]$ will be _____. **[J-Main 2022]**

CC0245

50. Low oxidation state of metals in their complexes are common when ligands: **[J-Main 2022]**

(A) have good π-accepting character (B) have good σ-donor character

(C) are havind good π-donating ability (D) are havind poor σ-donating ability

CC0246

51. The difference between spin only magnetic moment values of $[Co(H_2O)_6]Cl_2$ and $[Cr(H_2O)_6]Cl_3$ is ____. **[J-Main 2022]**

CC0247

52. Consider the following metal complexes : **[J-Main 2022]**

$[Co(NH_3)]^{3+}$

$[CoCl(NH_3)_5]^{2+}$

$[Co(CN)_6]^{3-}$

$[Co(NH_3)_5(H_2O)]^{3+}$

The spin-only magnetic moment value of the complex that absorbs light with shortest wavelength is B.M. (Nearest integer) **CC0248**

53. The correct order of energy of absorption for the following metal complexes is **[J-Main 2022]**

A : $[Ni(en)_3]^{2+}$, B : $[Ni(NH_3)_6]^{2+}$, C : $[Ni(H_2O)_6]^{2+}$

(A) C < B < A (B) B < C < A (C) C < A < B (D) A < C < B **CC0249**

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EXERCISE # JEE-ADVANCED

1. Spin only magnetic moment of the compound $Hg[Co(SCN)_4]$ is **[JEE 2004]**

(A) $\sqrt{3}$ (B) $\sqrt{15}$ (C) $\sqrt{24}$ (D) $\sqrt{8}$ **CC0181**

2. Which of the following pair is expected to exhibit same colour in solution? **[JEE 2005]**

(A) $VOCl_2$; $FeCl_2$ (B) $CuCl_2$; $VOCl_2$ (C) $MnCl_2$; $FeCl_2$ (D) $FeCl_2$; $CuCl_2$

CC0182

3. Which type of isomerism is shown by $Co(NH_3)_4Br_2Cl$? **[JEE 2005]**

(A) Geometrical and Ionisation

(B) Optical and Ionisation

(C) Geometrical and Optical

(D) Geometrical only

Question No. 8 to 10 (3 questions) **[JEE 2006]**

The coordination number of Ni^{2+} is 4.

$NiCl_2$ + KCN (excess) $\rightarrow$ A (cyano complex)

$NiCl_2$ + KCl (excess) $\rightarrow$ B (chloro complex) **CC0183**

4. The IUPAC name of A and B are

(A) Potassium tetracyanidonickelate (II), potassium tetrachloridonickelate (II)

(B) Tetracyanidopotassiumnickelate (II), teterachloridopotassiumnickelate (II)

(C) Tetracyanidornickel (II), tetrachloridonickel (II)

(D) Potassium tetracyanidonickel (II), potassium tetrachloridonickel (II) **CC0184**

5. Predict the magnetic nature of A and B.

(A) Both are diamagnetic.

(B) A is diamagnetic and B is paramagnetic with one unpaired electron.

(C) A is diamagnetic and B is paramagnetic with two unpaired electrons.

(D) Both are paramagnetic. **CC0185**

6. The hybridization of A and B are

(A) dsp^2, sp^3 (B) sp^3, sp^3 (C) dsp^2, dsp^2 (D) sp^3d^2, d^2sp^3

CC0186

7. If the bond length of CO bond in carbon monoxide is 1.128Å, then what is the value of CO bond length in $Fe(CO)_5$? **[JEE 2006]**

(A) 1.15Å (B) 1.128Å (C) 1.72Å (D) 1.118Å **CC0187**

8. Among the following metal carbonyls, the C–O bond order is lowest in **[JEE 2007]**

(A) $[Mn(CO)_6]^+$ (B) $[Fe(CO)_5]$ (C) $[Cr(CO)_6]$ (D) $[V(CO)_6]^-$

CC0188

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9. Match the complexes in Column I with their properties listed in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS. **[JEE 2007]**

Column I	Column II
(A) $[Co(NH_3)_4(H_2O)_2]Cl_2$	(P) Geometrical isomers
(B) $[Pt(NH_3)_2Cl_2]$	(Q) Paramagnetic
(C) $[Co(H_2O)_5Cl]Cl$	(R) Diamagnetic
(D) $[Ni(H_2O)_6]Cl_2$	(S) Metal ion with 2+ oxidation state

CC0189

10. Among the following, the coloured compound is **[JEE 2008]**

(A) CuCl (B) $K_3[Cu(CN)_4]$ (C) CuF_2 (D) $[Cu(CH_3CN)_4]BF_4$

CC0190

11. The IUPAC name of $[Ni(NH_3)_4][NiCl_4]$ is **[JEE 2008]**

(A) Tetrachloronickel (II)-tetraamminenickel (II) (B) Tetraamminenickel (II)-tetrachloronickel (II)

(C) Tetraamminenickel (II)-tetrachloronickelate (II) (D) Tetrachloronickel (II)-tetraamminenickelate (0)

CC0191

12. Both $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$ are diamagnetic. The hybridisations of nickel in these complexes, respectively, are **[JEE 2008]**

(A) sp^3, sp^3 (B) sp^3, dsp^2 (C) dsp^2, sp^3 (D) dsp^2, dsp^2 **CC0192**

13. **Statement-1 :** The geometrical isomers of the complex $[M(NH_3)_4Cl_2]$ are optically inactive.
Statement-2 : Both geometrical isomers of the complex $[M(NH_3)_4Cl_2]$ possess axis of symmetry.

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True **[JEE 2008]**

CC0193

14. **Statement-1 :** $[Fe(H_2O)_5NO]SO_4$ is paramagnetic **[JEE 2008]**
Statement-2 : The Fe in $[Fe(H_2O)_5NO]SO_4$ has three unpaired electrons.

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True **CC0194**

15. The spin only magnetic moment value (in Bohr magneton units) of $Cr(CO)_6$ is **[JEE 2009]**

(A) 0 (B) 2.84 (C) 4.90 (D) 5.92 **CC0195**

16. The compound(s) that exhibit(s) geometrical isomerism is (are) : **[JEE 2009]**

(A) $[Pt(en)Cl_2]$ (B) $[Pt(en)_2]Cl_2$ (C) $[Pt(en)_2Cl_2]Cl_2$ (D) $[Pt(NH_3)_2Cl_2]$

CC0196

17. The number of water molecule(s) directly bonded to the metal centre in $CuSO_4.5H_2O$ is. **[JEE 2009]**

CC0197

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18. The ionization isomer of $[Cr(H_2O)_4Cl(NO_2)]Cl$ is - **[JEE 2010]**

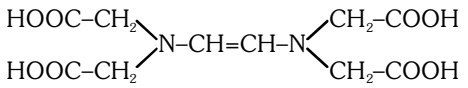
(A) $[Cr(H_2O)_4(O_2N)]Cl_2$ (B) $[Cr(H_2O)_4Cl_2](NO_2)$

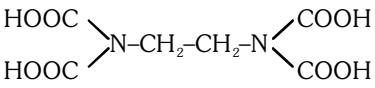
(C) $[Cr(H_2O)_4Cl(ONO)]Cl$ (D) $[Cr(H_2O)_4Cl_2(NO_2)].H_2O$ **CC0198**

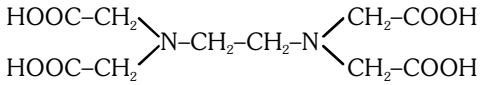
19. Total number of geometrical isomers for the complex $[RhCl(CO)(PPh_3)(NH_3)]$ is. **[JEE 2010]**

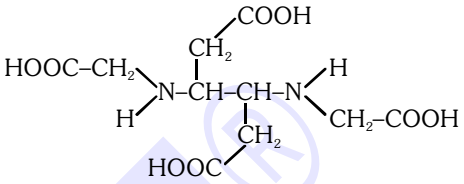
CC0199

20. The correct structure of ethylenediaminetetraacetic acid (EDTA) is – **[JEE 2010]**

(A) 

(B) 

(C) 

(D) 

CC0200

21. Geometrical shapes of the complexes formed by the reaction of Ni^{2+} with Cl^-, CN^- and H_2O respectively, are - **[JEE 2011]**

(A) octahedral, tetrahedral and square planar (B) tetrahedral, square planar and octahedral

(C) square planar, tetrahedral and octahedral (D) octahedral, square planar and octahedral

CC0201

22. Among the following complexes (**K–P**) **[JEE 2011]**

$K_3[Fe(CN)_6]$ **(K),** $[Co(NH_3)_6]Cl_3$ **(L),** $Na_3[Co(oxalate)_3]$ **(M),** $[Ni(H_2O)_6]Cl_2$ **(N),** $K_2[Pt(CN)_4]$ **(O)** and $[Zn(H_2O)_6](NO_3)_2$ **(P)**

The diamagnetic complex are -

(A) K, L, M, N (B) K, M, O, P (C) L, M, O, P (D) L, M, N, O

CC0202

23. The volume (in mL) of 0.1M $AgNO_3$ required for complete precipitation of chloride ions present in 30 mL of 0.01M solution of $[Cr(H_2O)_5Cl]Cl_2$, as silver chloride is close to. **[JEE 2011]**

CC0203

24. As per IUPAC nomenclature, the name of the complex $[Co(H_2O)_4(NH_3)_2]Cl_3$ is : **[JEE 2012]**

(A) Tetraaquadiaminecobalt(III) chloride (B) Tetraaquadiamminecobalt(III) chloride

(C) Diaminetetraaquacobalt(III) chloride (D) Diamminetetraaquacobalt(III) chloride

CC0204

25. The colour of light absorbed by an aqueous solution of $CuSO_4$ is - **[JEE 2012]**

(A) orange-red (B) blue-green (C) yellow (D) violet **CC0205**

26. $NiCl_2\{P(C_2H_5)_2(C_6H_5)\}_2$ exhibits temperature dependent magnetic behavior (paramagnetic/diamagnetic). The coordination geometries of Ni^{2+} in the paramagnetic and diamagnetic states are respectively :

[JEE 2012]

(A) tetrahedral and tetrahedral (B) square planar and square planar

(C) tetrahedral and square planar (D) square planar and tetrahedral **CC0206**

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27. Consider the following complex ions P, Q and R ,

P = $[FeF_6]^{3-}$, **Q** = $[V(H_2O)_6]^{2+}$ and **R** = $[Fe(H_2O)_6]^{2+}$

The **CORRECT** order of the complex ions, according to their spin-only magnetic moment values (in B.M.) is - **[JEE 2013]**

(A) R < Q < P (B) Q < R < P (C) R < P < Q (D) Q < P < R **CC0207**

28. $EDTA^{4-}$ is ethylenediaminetetraacetate ion. The total number of N–Co–O bond angles in $[Co(EDTA)]^{-1}$ complex ion is **[JEE 2013]**

CC0208

29. The pair(s) of coordination complex/ion exhibiting the same kind of isomerism is(are) - **[JEE 2013]**

(A) $[Cr(NH_3)_5Cl]Cl_2$ and $[Cr(NH_3)_4Cl_2]Cl$ (B) $[Co(NH_3)_4Cl_2]^+$ and $[Pt(NH_3)_2(H_2O)Cl]^+$

(C) $[CoBr_2Cl_2]^{2-}$ and $[PtBr_2Cl_2]^{2-}$ (D) $[Pt(NH_3)_3(NO_3)]$ Cl and $[Pt(NH_3)_3Cl]$ Br

CC0209

30. Match each coordination compound in List-I with an appropriate pair of characteristics from List-II and select the correct answer using the code given below the lists. **[JEE Adv. 2014]**

{en = $H_2NCH_2CH_2NH_2$ ' atomic numbers ; Ti = 22 ; Cr = 24 ; Co = 27 ; Pt = 78}

List-I	**List-II**
(P) $[Cr(NH_3)_4Cl_2]Cl$	(1) Paramagnetic and exhibits ionisation isomerism
(Q) $[Ti(H_2O)_5Cl](NO_3)_2$	(2) Dimagnetic and exhibits *cis-trans* isomerism
(R) $[Pt(en)(NH_3)Cl]NO_3$	(3) Paramagnetic and exhibits *cis-trans* isomerism
(S) $[Co(NH_3)_4(NO_3)_2]NO_3$	(4) Dimagnetic and exhibits ionisation isomerism

Code :

	P	**Q**	**R**	**S**		**P**	**Q**	**R**	**S**
(A)	4	2	3	1	(B)	3	1	4	2
(C)	2	1	3	4	(D)	1	3	4	2

CC0210

31. A list of species having the formula XZ_4 is given below : **[JEE Adv. 2014]**

XeF_4, SF_4, SiF_4, BF_4^-, BrF_4^-, $[Cu(NH_3)_4]^{2+}$, $[FeCl_4]^{2-}$, $[CoCl_4]^{2-}$ and $[PtCl_4]^{2-}$.

Defining shape on the basis of the location of X and Z atoms, the total number of species having a square planar shape is **CC0211**

SUBJECTIVE :

32. Draw the structures of $[Co(NH_3)_6]^{3+}$, $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$. Write the hybridisation of atomic orbitals of the transition metal in each case. **[JEE 2000]**

CC0212

33. A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has been isolated in two forms A and B. The form A reacts with $AgNO_3$ to give a white precipitate readily soluble in dilute aqueous ammonia, whereas B gives a pale yellow precipitate soluble in concentrated ammonia. Write the formula of A and B and state the hybridisation of chromium in each. Calculate their magnetic moments (spin-only value).

[JEE 2001]

CC0213

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34. Deduce the structures of $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ considering the hybridisation of the metal ion. Calculate the magnetic moment (spin only) of the species. **[JEE 2002]**
CC0214

35. Write the IUPAC name of the compound $K_2[Cr(NO)(CN)_4(NH_3)]$. Spin magnetic moment of the complex $\mu = 1.73$ BM. Give the structure of anion. **[JEE 2003]**
CC0215

36. $NiCl_2$ in the presence of dimethyl glyoxime (DMG) gives a complex which precipitates in the presence of NH_4OH, giving a bright red colour. **[JEE 2004]**
(a) Draw its structure and show H-bonding
(b) Give oxidation state of Ni and its hybridisation
(c) Predict whether it is paramagnetic or diamagnetic **CC0216**

37. For the octahedral complexes of Fe^{3+} in SCN^- (thiocyanato-S) and in CN^- ligand environments, the difference between the spin only magnetic moments in Bohr magnetons (when approximated to the nearest integer) is : **[Atomic number of Fe = 26]** **[JEE Ad. 2015]**
CC0217

38. In the complex acetylbromidodicarbonylbis(triethylphosphine)iron(II), the number of Fe–C bond(s) is-
[JEE Ad. 2015]
CC0218

39. Among the complex ions, $[Co(NH_2\text{-}CH_2\text{-}CH_2\text{-}NH_2)_2Cl_2]^+$, $[CrCl_2(C_2O_4)_2]^{3-}$, $[Fe(H_2O)_4(OH)_2]^+$, $[Fe(NH_3)_2(CN)_4]^-$, $[Co(NH_2\text{-}CH_2\text{-}CH_2\text{-}NH_2)_2(NH_3)Cl]^{2+}$ and $[Co(NH_3)_4(H_2O)Cl]^{2+}$, the number of complex ion(s) that show(s) *cis-trans* isomerism is - **[JEE Ad. 2015]**
CC0219

40. Among $[Ni(CO)_4]$, $[NiCl_4]^{2-}$, $[Co(NH_3)_4Cl_2]Cl$, $Na_3[CoF_6]$, Na_2O_2 and CsO_2, the total number of paramagnetic compounds is - **[JEE Ad. 2016]**
(A) 2 (B) 3 (C) 4 (D) 5 **CC0220**

41. The number of geometric isomers possible for the complex $[CoL_2Cl_2]^-$ ($L = H_2NCH_2CH_2O^-$) is
[JEE Ad. 2016]
CC0221

42. The geometries of the ammonia complexes of N^{2+}, Pt^{2+} and Zn^{2+} respectively, are :**[JEE Ad. 2016]**
(A) octahedral, square planar and tetrahederal (B) square planar, octahederal and tetrahederal
(C) tetrahederal, square planar and octahederal (D) octahederal, tetrahederal and square planar
CC0222

43. Addition of excess aqueous ammonia to a pink coloured aqueous solution of $MCl_2 . 6H_2O$ (X) and NH_4Cl gives an octahedral complex Y in the presence of air. In aqueous solution, complex Y behaves as 1 : 3 electrolyte. The reaction of X with excess HCl at room temperature results in the formation of a blue coloured complex Z. The calculated spin only magnetic moment of X and Z is 3.87 B.M., whereas it is zero for complex Y. **[JEE Ad. 2017]**
Among the following options, which statements is(are) **CORRECT** ?
(A) The hybridization of the central metal ion in Y is d^2sp^3
(B) Z is tetrahedral complex
(C) Addition of silver nitrate to Y gives only two equivalents of silver chloride
(D) When X and Z are in equilibrium at 0°C, the colour of the solution is pink **CC0223**

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44. The correct statement(s) regarding the binary transition metal carbonyl compounds is (are)

(Atomic numbers : Fe = 26, Ni = 28) **[JEE Ad. 2018]**

(A) Total number of valence shell electrons at metal centre in $Fe(CO)_5$ or $Ni(CO)_4$ is 16

(B) These are predominantly low spin in nature

(C) Metal - carbon bond strengthens when the oxidation state of the metal is lowered

(D) The carbonyl C–O bond weakens when the oxidation state of the metal is increased

CC0224

45. The correct option(s) regarding the complex $[Co(en)(NH_3)_3(H_2O)]^{3+}$:- **[JEE Ad. 2018]**

(en = $H_2NCH_2CH_2NH_2$) is (are)

(A) It has two geometrical isomers

(B) It will have three geometrical isomers if bidentate 'en' is replaced by two cyanide ligands

(C) It is paramagnetic

(D) It absorbs light at longer wavelength as compared to $[Co(en)(NH_3)_4]^{3+}$ **CC0225**

46. Match each set of hybrid orbitals from LIST-I with complex (es) given in LIST-II.

LIST-I	**LIST-II**
P. dsp^2	1. $[FeF_6]^{4-}$
Q. sp^3	2. $[Ti(H_2O)_3Cl_3]$
R. sp^3d^2	3. $[Cr(NH_3)_6]^{3+}$
S. d^2sp^3	4. $[FeCl_4]^{2-}$
	5. $Ni(CO)_4$
	6. $[Ni(CN)_4]^{2-}$

[JEE Ad. 2018]

The correct option is

(A) P → 5; Q → 4,6; R → 2,3; S → 1

(B) P → 5,6; Q → 4; R → 3; S → 1,2

(C) P → 6; Q → 4,5; R → 1; S → 2,3

(D) P → 4,6; Q → 5,6; R → 1,2; S → 3

CC0226

47. Among the species given below, the total number of diamagnetic species is____.

H atom, NO_2 monomer, O_2^- (superoxide), dimeric sulphur in vapour phase, **[JEE Ad. 2018]**

Mn_3O_4, $(NH_4)_2[FeCl_4]$, $(NH_4)_2[NiCl_4]$, K_2MnO_4, K_2CrO_4 **CC0227**

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48. Total number of *cis* N–Mn–Cl bond angles (that is, Mn–N and Mn–Cl bonds in *cis* positions) present in a molecule of *cis*-$[Mn(en)_2Cl_2]$ complex is ____ (*en* = $NH_2CH_2CH_2NH_2$) **[JEE Ad. 2019]**

CC0228

49. Choose the correct statement(s) among the following : **[JEE Ad. 2020]**

(A) $[FeCl_4]^-$ has tetrahedral geometry.

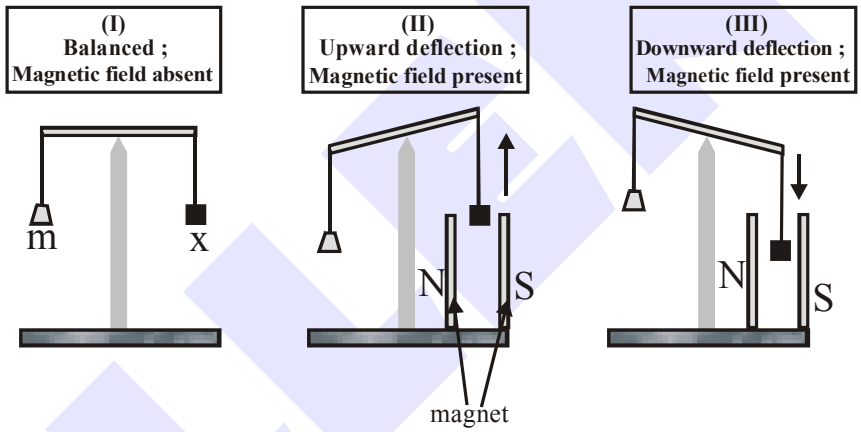
(B) $[Co(en)(NH_3)2Cl_2]^+$ has 2 geometrical isomers.

(C) $[FeCl_4]^-$ has higher spin-only magnetic moment than $[Co(en)(NH_3)2Cl_2]^+$.

(D) The cobalt ion in $[Co(en)(NH_3)2Cl_2]$ has sp^3d^2 hybridization.

CC0230

50. In an experiment, m grams of a compound X (gas/liquid/solid) taken in a container is loaded in a balance as shown in figure **I** below. In the presence of a magnetic field, the pan with **X** is either deflected upwards (figure **II**), or deflected downwards (figure **III**), depending on the compound **X**. Identify the correct statement(s) **[JEE Ad. 2020]**



(A) If **X** is $H_2O(\ell)$, deflection of the pan is upwards.

(B) If **X** is $K_4[Fe(CN)_6](s)$, deflection of the pan is upwards.

(C) If **X** is $O_2(g)$, deflection of the pan is downwards.

(D) If **X** is $C_6H_6(\ell)$, deflection of the pan is downwards.

CC0231

51. The calculated spin only magnetic moments of $[Cr(NH_3)_6]^{3+}$ and $[CuF_6]^{3-}$ in BM, respectively, are (Atomic numbers of Cr and Cu are 24 and 29, respectively) **[JEE ADV. 2021]**

(A) 3.87 and 2.84 (B) 4.90 and 1.73 (C) 3.87 and 1.73 (D) 4.90 and 2.84

CC0240

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52. The pair(s) of complexes wherein both exhibit tetrahedral geometry is(are) **[JEE ADV. 2021]**

(Note: py = pyridine

Given: Atomic numbers of Fe, Co, Ni and Cu are 26, 27, 28 and 29, respectively)

(A) $[FeCl_4]^-$ and $[Fe(CO)_4]^{2-}$

(B) $[Co(CO)_4]^-$ and $[CoCl_4]^{2-}$

(C) $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$

(D) $[Cu(py)_4]^+$ and $[Cu(CN)_4]^{3-}$

CC0241

53. The total number of possible isomers for $[Pt(NH_3)_4Cl_2]Br_2$ is _____ **[JEE ADV. 2021]**

CC0242

54. **LIST-I** contains metal species and **LIST-II** contains their properties. **[JEE ADV. 2022]**

LIST-I	**LIST-II**
(I) $[Cr(CN)_6]^{4-}$	(P) t_{2g} orbitals contain 4 electrons
(II) $[RuCl_6]^{2-}$	(Q) μ(spin-only) = 4.9 BM
(III) $[Cr(H_2O)_6]^{2+}$	(R) low spin complex ion
(IV) $[Fe(H_2O)_6]^{2+}$	(S) metal ion in 4+ oxidation state
	(T) d^4 species

[Given : Atomic number of Cr = 24, Ru = 44, Fe = 26]

Metal each metal species in LIST-I with their properties in LIST-II, and choose the correct option.

(A) I→R, T ; II→P, S ; III→Q, T ; IV→P, Q

(B) I→R, S ; II→P, T ; III→P, Q ; IV→Q, T

(C) I→P, R ; II→R, S ; III→R, T ; IV→P, T

(D) I→Q, T ; II→S, T ; III→P, T ; IV→Q, R

CC0250

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ANSWERS KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	B	C	C	C	D	B	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	A	B	A	C	C	D	A	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	C	B	C	B	B	A	D	A	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	D	B	C	D	C	C	B	C	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	B	C	D	D	D	D	C	B	D	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A	B	A	D	B	D	A	C	B	D
Que.	61	62	63	64	65	66	67	68	69	
Ans.	B	C	C	A	C	C	A	D	A	

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	A,B,D	A,B,C,D	A,B,D	A,B	B,C,D	A,B,C,D	A,B,C	B,C,D	A,B,C,D
Que.	11	12	13	14						
Ans.	B,D	C,D	C,D	B,D						

EXERCISE # S-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	9	36	9	2	2	52	5	2

EXERCISE # S-2

Que.	1				2			
Ans.	(A)-R,	(B)-R	(C)-P	(D)-Q	(A)-P,Q,S	(B)-Q,R,S	(C)-P	(D)-P,Q
Que.	3							
Ans.	(A)-P,S	(B)-T	(C)-Q,R,T	(D)-P,S				

Que.	4	5	6	7	8	9	10	11	12	13
Ans.	B	C	D	D	C	A	C	D	D	A
Que.	14	15	16	17	18					
Ans.	D	B	B	D	D					

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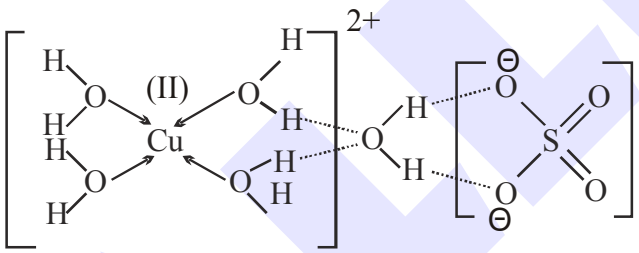
EXERCISE # J-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	1	3	3	4	3	4	4	4
Que.	11	12	13	14	15	16	17	18	19	20
	1	3	3	3	2	4	4	1	3	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	3	3	1	2	2	1	2	2	1
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	3	4	3	2	2	4	1	1	4	26.60-27.70
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	0	2	2	2	2	4	A	B	3	A
Que.	51	52	53							
Ans.	0	0	A							

EXERCISE # J-ADVANCED

Que.	1	2	3	4	5	6	7	8		
Ans.	B	B	A	A	C	A	A	D		
Que.	9						10	11	12	13
Ans.	(A)-P,Q,S ; (B)-PR,S ; (C)-Q,S ; (D)-Q,S						C	C	B	B
Que.	14	15	16							
Ans.	A	A	C, D							

17.



Que.	18	19	20	21	22	23	24	25	26	27
Ans.	B	3	C	B	C	6	D	A	C	B
Que.	28	29	30	31						
Ans.	8	B, D	B	4						

32. d^2sp^3 , dsp^2 and sp^3

33. A $\rightarrow$ $[Cr(NH_3)_4ClBr]Cl$

B $\rightarrow$ $[Cr(NH_3)_4Cl_2]Br$

In both Cr is d^2 sp^3 hybridised and magnetic moment is $\sqrt{15}$ BM

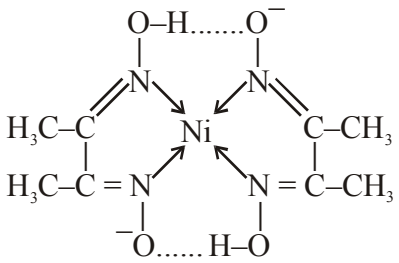
34. $[NiCl_4]^{2-} \rightarrow sp^3$, $\sqrt{8}$ BM

$[Ni(CN)_4]^{2-} \rightarrow dsp^2$, 0

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35. Potassium amminetetracyanidonitrosoniumchromate(I) $\rightarrow d^2sp^3$, octahedral

36.



dsp^2 , Ni^{2+} , diamagnetic

Que.	37	38	39	40	41	42	43	
Ans.	4	3	6	B	5	A	A,B,D	

43. $\overset{II}{[Co(H_2O)_6]Cl_2} \xrightarrow[Air\ /\ O_2]{Excess\ NH_4OH\ /\ NH_4Cl} \overset{III}{[Co(NH_3)_6]Cl_3}$

Pink **(X)** **(Y)**

$[Co(H_2O)_6]^{2+} + 4Cl^- \longrightarrow [CoCl_4]^{2-}$

(X) (excess) **(Z)** blue colour

(A) Hybridisation of (Y) is d^2sp^3 as NH_3 is strong field ligand

(B) $[CoCl_4]^{2-}$ have sp^3 hybridisation as Cl^- is weak field ligand

(C) $\overset{III}{[Co(NH_3)_6]Cl_3} + 3AgNO_3(aq.) \rightarrow 3AgCl$

(Y)

(D) $\underset{(Blue)}{[CoCl_4]^{2-}} + 6H_2O \rightleftharpoons \underset{(Pink)}{[Co(H_2O)_6]^{2+}} + 4Cl^-$ ΔH = (–)ve (exothermic)

When ice is added to the solution the equilibrium shifts right hence pink colour will remain predominant

So, correct answer is (A,B& D)

Que.	44	45	46	47	48	49	50	51	52	53
Ans.	B,C	A,B,D	C	1	6.00	A,C	A,B,C	A	A,B,D	6
Que.	54									
Ans.	A									

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